

SULTAN AL SHAFIAN

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EDUCATION

PhD Candidate in Interdisciplinary Engineering (ABD), Kennesaw State University, GA, USA Ongoing

Research Focus: Smart Infrastructure

Ph.D. Research Topic: Disaster Reconnaissance

Project: Advancing Spontaneous Volunteer Coordination and Guidance for Effective Disaster Response

NSF Award No. 2346936

M.Sc. in Civil Engineering, Islamic University of Technology

2023

Thesis: Deformation and Stress Analyses of the Ground During the Construction of Multipurpose Road Tunnel Under Karnaphuli River

B.Sc. in Civil Engineering, Islamic University of Technology

2015

Thesis: Investigation of Concrete Properties with Partial Replacement of Polypropylene Plastic

HONORS, AWARDS & GRANTS

External Grants

National Science Foundation (NSF) I-Corps Teams Program (\$50,000)

Entrepreneurial Lead (EL) for AutoMap Subsurface, a GPR-based technology for automated subsurface utility mapping and visualization, 2026 - 2027

Affordable Learning Georgia (\$10,000)

Developer for Augmented Reality Lab for Geotechnical Engineering, 2025

Honors and Awards

Best Student Paper Award

SPCEET Author's Expo, Kennesaw State University, 2025

Graduate Travel Award

Graduate College, Kennesaw State University, 2024 & 2025

Runner Up, Poster Presentation

CENNOVATION, Islamic University of Technology, 2020

PUBLICATIONS

Journals

1. Al Shafian, S., He, C., O'Neal, K., Xu, H., Khalid, A., & Hu, D. (2026). **AFTERMAP: A FEMA-Aligned Dataset for Damaged Structure Segmentation in UAV Imagery**. *Buildings*, 16(15), 2943. <https://doi.org/10.3390/buildings16152943>

2. McNeil, S., Amekudzi-Kennedy, A., Torres-Machi, C., Al Shafian, S., Bukenya, I., Chukwu, J. O., Contreras-Jara, M., Cuadra, M., Cui, B., Kandel, M., Narh, J. N., Shehata, L., Teixeira, P., Tetteh, N. O., & Tsogt-Ochir, N. (2026). **A Guide to Using Artificial Intelligence to Enhance Transportation Asset-Management Plans**. *Transportation Research Record*. <https://doi.org/10.1177/03611981261465428>

3. Al Shafian, S., He, C., & Hu, D. (2025). **DamageScope: An integrated pipeline for building damage segmentation, geospatial mapping, and interactive web-based visualization**. *Remote Sensing*, 17(13), 2267. <https://doi.org/10.3390/rs17132267>

4. Al Shafian, S., & Hu, D. (2024). **Integrating machine learning and remote sensing in disaster management: A decadal review of post-disaster building damage assessment**. *Buildings*, 14(8), 2344. <https://doi.org/10.3390/buildings14082344>

5. Ahmed, T., Chowdhury, L. F., Sobhani, M. G., Islam, A. S. M. M., **Al Shafian, S.**, & Majumder, M. (2023). **Enhancing urban mobility: Exploring the potential of exclusive motorcycle lanes using VISSIM.** *Journal of Transportation Technologies*, 13(4), 644–656.
6. Imtiyaz, M. N., Emtiaz, M., **Al Shafian, S.**, & Shahin, H. M. (2023). **A finite element approach to investigate the deformation behaviour in deep excavation for TBM launching shaft.** *International Journal of Civil Infrastructure*, 6, 36–40.
7. Ahmad, K. I., Abrar, M., **Al Shafian, S.**, & Shahin, H. M. (2018). **Correlation among the soil parameters of the Karnaphuli River Tunnel Project.** *International Journal of GEOMATE*, 15(48), 86–90.

Conference Publications

1. **Al Shafian, S.**, Karu, J. & Hu, D. (2027). **A SUMO–Unity Framework for First-Responder Training and Area Familiarization in Post-Disaster Environments.** Accepted for presentation at *ASCE Conference*, Philadelphia, United States, March 1–5, 2027.
2. **Al Shafian, S.**, Rhythy, T. R., Gao, S., Xu, H., Khalid, A., & Hu, D. (2026). **Post-disaster building damage assessment using drone imagery and deep learning.** *Construction Research Congress 2026*, San Antonio, TX, United States, March 18–21, 2026.
3. **Al Shafian, S.** & Hu, D. (2026). **Immersive virtual reality training for first responders using photogrammetry based 3D models of disaster environments.** *Construction Research Congress 2026*, San Antonio, TX, United States, March 18–21, 2026.
4. **Al Shafian, S.**, Xu, H., Khalid, A., & Hu, D. (2026). **Simulating real-world damage environments for interactive virtual reality training via 3D Gaussian splatting.** Accepted for presentation at *IEEE 4th International Conference on Intelligent Reality (ICIR 2025)*, Northeastern University, Boston, Massachusetts, USA, November 16–18, 2025. [10.1109/ICIR68135.2025.11361594](https://doi.org/10.1109/ICIR68135.2025.11361594)
5. **Al Shafian, S.**, Hu, D., & Alam, M. S. (2025). **Immersive simulations and AI-enhanced approaches for enhancing post-disaster response.** *ASCE International Conference on Computing in Civil Engineering 2025*, New Orleans, LA, United States, May 11–14, 2025. <https://doi.org/10.1061/9780784486443.006>
6. **Al Shafian, S.**, & Hu, D. (2025). **Integrating drone imaging and deep learning with augmented reality for disaster damage assessment and response.** *ASCE International Conference on Computing in Civil Engineering 2025*, New Orleans, LA, United States, May 11–14, 2025.
7. Alam, M. S., **Al Shafian, S.**, Kaplan, A., Gooding, A., & Hu, D. (2025). **Developing a test bed to enhance ground-penetrating radar data preparation and analysis.** *IEEE International Radar Conference*, Atlanta, GA, United States, May 5–9, 2025. <https://doi.org/10.1109/RADAR52380.2025.11031574>
8. **Al Shafian, S.**, Hu, D., Kwon, J., & Kaplan, A. (2025). **Augmented reality for teaching rebar configurations: Improving comprehension and student engagement.** *ASEE Southeast Conference 2025*, Mississippi State University, March 9–11, 2025. <https://peer.asee.org/54145>
9. **Al Shafian, S.**, Hu, D., & Yu, W. (2024). **Deep learning-enhanced crack detection for tunnel inspection.** *International Conference on Transportation and Development 2024*. <https://doi.org/10.1061/9780784485514.06>
10. **Al Shafian, S.**, & Hu, D. (2024). **Harnessing synthetic image datasets for enhanced scene understanding in construction work zones.** *International Conference on Transportation and Development 2024*. <https://doi.org/10.1061/9780784485514.063>
11. **Al Shafian, S.**, Hu, D., Li, Y., & Adhikari, S. (2024). **Improving construction site safety by incident reporting through virtual reality.** *ASEE Southeastern Section Meeting 2024*. <https://doi.org/10.18260/1-2--45535>
12. Hu, D., & **Al Shafian, S.** (2024). **Segmentation and tracking of moving objects on dynamic construction sites.** In *Construction Research Congress 2024* (pp. 60–69).
13. **Al Shafian, S.**, Zhenqi, J., Shaheduzzaman, M., Nazrul, F. B., Hridoy, D. N., & Hossain, M. (2023). **Analysis of the load–displacement behaviour of bored piles using different soil constitutive models for Chittagong**

soil. *8th World Congress on Civil, Structural, and Environmental Engineering (CSEE'23)*, Lisbon, Portugal, March 29–31, 2023.

14. **Al Shafian, S.**, Talha, S. A., & Hossain, M. (2022). **Understanding the effect of degree of saturation on compressive strength behaviours for reconstituted clayey soil of Dhaka.** *11th International Conference on Geotechnique, Construction, Materials and Environment*, Kyoto, Japan, November 3–5, 2022.

15. Emtiaz, M., Al Azad, A. S. M. A., Shahin, H. M., & **Al Shafian, S.** (2017). **Numerical analysis of a reinforced concrete slab–column connection subjected to lateral and vertical loading.** In *Proceedings of the International Multi-Conference of Engineers and Computer Scientists 2017* (Vol. II), Hong Kong, March 15–17, 2017.

16. Islam, M. J., Sarwar, N., & **Al Shafian, S.** (2015). **An investigation of concrete properties with polypropylene (PP) as a partial replacement of coarse aggregate.** *International Conference on Recent Innovation in Civil Engineering for Sustainable Development (IICSD-2015)*.

17. Islam, M. J., **Al Shafian, S.**, & Sarwar, N. (2015). **Fresh and hardened properties of concrete with polypropylene.** *1st International Conference on Advances in Civil Infrastructure and Construction Materials (CICM)*, Dhaka, Bangladesh, December 13–14, 2015.

RESEARCH EXPERIENCE

Graduate Research Assistant
Kennesaw State University

August 2023 - Present
Kennesaw, Georgia, USA

- Conducted rapid post-disaster field reconnaissance following Hurricane Helene to document storm impacts, infrastructure damage, and site conditions.
- Developing AFTERMAP, a UAV-based post-disaster imagery dataset and deep-learning framework for building-damage segmentation and rapid damage assessment across multiple extreme-event scenarios.
- Developing DamageScope, an integrated framework combining AI-based damage detection, GIS-based geospatial mapping, and interactive visualization to support rapid post-disaster situational awareness.
- Conducting and supporting RTK-GNSS and total-station surveying, UAV-based mapping and photogrammetry, and processing of point-cloud and remote-sensing data for field and geospatial investigations.
- Performing GIS, spatial, satellite-imagery, and machine-learning analyses to identify, map, and characterize disaster-related damage and changes in the built environment.
- Supporting field campaign planning and coordination, research proposal development, and supervision of undergraduate researchers.

Research Assistant
Islamic University of Technology

January 2016 - September 2017
Gazipur, Bangladesh

- Performed geotechnical laboratory testing of soils, including sample preparation, testing, data interpretation, and preparation of technical reports.
- Conducted numerical analyses of slopes, ground deformation, and underground structures and tunnel systems to evaluate stability and soil–structure response.
- Assisted with laboratory equipment calibration, experimental planning, and research data management.
- Supported and supervised undergraduate students in laboratory testing, data analysis, and geotechnical research activities.

TEACHING EXPERIENCE

Adjunct Lecturer
Presidency University

January 2022 - December 2022
Dhaka, Bangladesh

- Courses: Introduction to programming, Principles of Geotechnical Engineering, Reinforced Concrete Design Sessional, Building Services

- Courses: HUM 285- Engineering, Engineering Drawing

ACADEMIC AND VOLUNTEER SERVICE

Reviewer, *Geofluids*, Wiley's Forward Series, 2024

Reviewer, *Automation in Construction*, Elsevier, 2024

Guest Lecturer, *Tunneling Short Course*, Islamic University of Technology, 2020

Speaker, *Industrial Training*, Islamic University of Technology, 2019

INDUSTRY EXPERIENCE

Tunnel Design Engineer

ACE Consultants Limited

January 2022 - July 2023

MRT Line 5 (South)

- Analyzed critical sections along the alignment and supported the design of underground structures for the MRT Line 5 project.
- Reviewed applicable design codes and prepared engineering design methodologies and technical documentation.
- Performed numerical modeling and soil-structure interaction analysis using PLAXIS 2D to evaluate ground settlement and potential impacts on adjacent infrastructure.

Assistant Tunnel Engineer

China Communications Construction Company Ltd (CCCC)

June 2018 - December 2021

Karnaphuli Tunnel Project

- Supervised bathymetric and land-filling surveys, including field coordination, survey-data review, and preparation of engineering documentation.
- Conducted and monitored surface-settlement and tunneling measurements to evaluate ground response during construction.
- Supported field surveying and geotechnical monitoring activities for the Karnaphuli Tunnel Project and integrated field observations into technical reports.
- Coordinated construction-site data collection and progress monitoring across large-scale infrastructure works.

Geotechnical Engineer

Prosoil Foundation and Consultants

October 2017 - May 2018

Dhaka, Bangladesh

- Conducted and supported geotechnical field and laboratory investigations, including Dynamic Cone Penetration (DCP) testing, plate-load testing, crosshole sonic logging (CSL) testing, consolidation, and direct shear testing
- Assisted with subsurface characterization, field data collection, interpretation of test results, and preparation of geotechnical reports.
- Coordinated field activities and technical discussions with clients and project teams.

TRAINING AND WORKSHOPS

- Statement of Accomplishment on “**Limit equilibrium slope stability analysis using Slide2 and Slide3**” by Ingeoexpert
- Statement of Accomplishment on “**Structural Health Monitoring**” by Ingeoexpert
- Statement of Accomplishment on “**Tunnel Design with PLAXIS**” by Ingeoexpert
- Two-week training in Public Works Department, Dhaka, Bangladesh
- Training program at Engineering Staff College, Bangladesh on STAAD.PRO

LEADERSHIP EXPERIENCE

- Vice-President, KSU ITE Student Chapter (April 2025-Present)
- Vice-President, Bangladeshi Students and Teachers Association (BSTA) at KSU (January 2024 - December 2024)
- Internship Coordinator for Undergraduate Student Groups, Sthapathi Consultants (2021)
- General Secretary, CENNOVATION 2015, National Civil Engineering Student Competition, Bangladesh

REFERENCES

Dr. Da Hu

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Department of Civil and Environmental Engineering
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Marietta, Georgia, USA
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Dr. Tien Yee

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